What is Optimization

> Optimization = finding the **best solution among all possible solutions**

Simple daily examples

- fastest route to university
- cheapest ride
- best study schedule

Key Components of Optimization

1️⃣ Decision Variables

Things we can **control or decide**.

Example

`x = number of lemonade cups`

2️⃣ Objective Function

What we want to **maximize or minimize**

Examples

- maximize profit
- minimize cost
- minimize time

General form

```
max / min   f(x1 , x2 , ... , xn)
```

3️⃣ Constraints

Limitations or restrictions.

Examples

- limited resources
- time limits
- production capacity

General form

```
g1(x1,...xn) ≤ b1
g2(x1,...xn) ≤ b2
```

Example — Lemonade Problem

Decision Variable

```
x = number of lemonade cups
```

Selling price = 5
Cost per cup = 2

Profit

```
Profit = 5x − 2x
Profit = 3x
```

Objective Function

```
max Z = 3x
```

Constraints

```
0.1x ≤ 10     (sugar)
0.5x ≤ 50     (lemons)
0.02x ≤ 2     (time)
x ≥ 0
```

Types of Optimization Problems

Optimization problems are commonly classified into three main types.

Linear Programming (LP)

> Linear Programming

A **Linear Programming problem** is an optimization problem where:

- the **objective function is linear**
- the **constraints are linear**
- the **decision variables are continuous**

Example

```
max 3x₁ + 5x₂
```

subject to

```
x₁ + 2x₂ ≤ 10
2x₁ + x₂ ≤ 12
x₁ , x₂ ≥ 0
```

LP problems are usually **easier to solve** and always have a **global optimal solution**.

Applications

- production planning
- resource allocation
- portfolio optimization
- diet problems

➡️ For a detailed explanation see: Linear Programming

Nonlinear Programming (NLP)

> Nonlinear Programming

A **Nonlinear Programming problem** occurs when the **objective function or constraints are nonlinear**.

Example

```
min x₁² + x₂²
```

subject to

```
x₁ + x₂ = 1
x₁ , x₂ ≥ 0
```

NLP problems are **more complex** because they may contain **multiple local optimum solutions**.

Applications

- machine learning
- robotics
- chemical process optimization

➡️ For a deeper explanation see: Nonlinear Programming

Integer Programming (IP)

> Integer Programming

In **Integer Programming**, the decision variables must take **integer values**.

Example

```
xᵢ = 1   project selected
xᵢ = 0   project not selected
```

These are called **binary decision variables**.

Applications

- job scheduling
- facility location
- network design

- project selection

Integer programming problems are **harder to solve** because solutions must be integers.

➡️ For full details see: Integer Programming

Comparison

Type	Function	Variables	Difficulty
LP	Linear	Continuous	Easy
NLP	Nonlinear	Continuous	Medium
IP	Linear / Nonlinear	Integer	Hard

Relationship Between Them

```
Optimization
│
├── Linear Programming
├── Nonlinear Programming
└── Integer Programming
```

All three are **different ways of modeling optimization problems** depending on the structure of the system.

Comparison of LP, NLP, and IP

Aspect	LP	NLP	IP
Linearity	Linear	Nonlinear	Linear / Nonlinear
Variables	Continuous	Continuous	Integer
Difficulty	Easy	Medium	Hard
Solve Time	Fast	Medium	Can be slow
Global Optimum	Guaranteed	Maybe not	Guaranteed*

Continuous vs Discrete Variables

Type	Meaning	Example
Continuous	any real value	2.5 liters
Discrete	integers only	3 workers

Continuous examples

- production volume
- temperature
- investment

Discrete examples

- number of workers
- number of flights
- yes/no decisions

Steps to Formulate an Optimization Problem

1. Identify **decision variables**
2. Define **objective function**
3. Write **constraints**
4. Check **mathematical correctness**

> **Important**
>
> Hardest part = **formulating the problem correctly**

Common Modeling Mistakes

Mistake	Example
Forgetting non-negativity	$x \geq 0$
Inconsistent units	kg vs pounds
Missing constraints	unrealistic solution
Overcomplicated model	hard to solve

> **Tip**
>
> A good model should be **simple but realistic**

Real Applications

Google Search

Goal

```
maximize user satisfaction
```

Constraint

```
response time < 0.5 seconds
```

Amazon Delivery

Objective

```
minimize travel distance
```

Constraints

- truck capacity
- delivery time
- traffic

Problem name

```
Vehicle Routing Problem (VRP)
```

Netflix Recommendations

Goal

```
maximize viewing time
```

Constraints

- user preferences
- available content

Optimization in AI

Training a neural network is an **optimization problem**.

Example

```
min L(θ)
```

Where

```
θ = model parameters
L = loss function
```

Common algorithm

```
Gradient Descent
```

Key Ideas

Optimization problems always include:

- Decision Variables
- Objective Function
- Constraints

Main types:

- Linear Programming (LP)
- Nonlinear Programming (NLP)
- Integer Programming (IP)

Concept Map

```
Optimization
│
├── Decision Variables
├── Objective Function
└── Constraints
```

Optimization Types Map

```
Optimization
│
├── Linear Programming
├── Nonlinear Programming
└── Integer Programming
```